

Makis Lam

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Available May. - Aug. 2026 | Incoming MDE @ Formlabs

SKILLS

CAD & Analysis: SOLIDWORKS, CATIA (V5/V6), Inventor, AutoCAD, Ansys, Abaqus, 3DX, ENOVIA, Vault.

Manufacturing: 3D Printing, Sheet Metal, CNC Machining, GD&T, Tolerance Stackup, Validation & Reliability.

Design & Software: Python, C++, MATLAB, Simulink, Arduino, First Principles Analysis, DFMA, DFMEA, V&V.

EXPERIENCE

SMART-Lab

Sept. 2025 - Present

Robotics Undergraduate Research Assistant

Waterloo, ON

- Designed a magnetic rotating end effector with a 40:1 strain wave gear for a 6-axis robotic arm, using inverse kinematics in Python to precisely orient and position microrobots without linkage deadlocks.

Tesla

May - Aug. 2025

Robotics Hardware Engineer - Optimus

Palo Alto, CA

- Spearheaded calibration repeatability and accuracy validation by developing metrology study procedures (DOE), executing tests, and performing statistical analysis with Minitab to reduce assembly time by 15 minutes.
- Analyzed mainline operations using PFMEA to improve operator ergonomics and balance station workloads, reducing cycle time and enabling a transition to mass production with a significant increase in unit output.
- Engineered novel processes for installing robotic limbs and components, ensuring safety compliance, optimizing design for assembly (DFA), and enabling additional tasks during position-controlled or autonomous operation.
- Improved bot accuracy and kinematic understanding by designing dimensional measurement strategies and conducting a first-principles analysis of end-effector error sources, identifying key factors for better controllability.
- Designed poka-yoke jigs and fixtures in CATIA V6 to ensure repeatable and error-proof assembly processes.
- Triaged and resolved firmware and hardware issues across actuators and sensors using PCAN Explorer and CANape, and developed Python scripts to validate solutions, supporting production and meeting manufacturing takt times.

Formlabs

Sept. 2024 - Dec. 2024

Settings Optimization Engineer (R&D Engineer) - SLA Printers

Somerville, MA

- Optimized and identified failure modes in 6+ resins by a first-principles tuning method, participating in design reviews to inform decisions and drive development of SLA printers, owning ~3% of settings by volume in Form 4L.
- Investigated flowmark print surface artifacts, identifying root causes, conditions for prominence, and mitigation strategies, and developed a Python script to manipulate the print process in Form 4, eliminating defects.
- Led and increased efficiency of reliability testing on new-generation LCD panels by 57% by proposing testing with smaller layer heights, collecting data to verify the approach, and successfully implementing the change.
- Employed metrology tools to compare dimensions of printed parts supporting printer R&D and cost-down efforts.

Airbus

Jan. - Apr. 2024

Integration Engineer - A220 Cabin

Mirabel, QC

- Delivered 15+ customer response (CRT) presentations by conducting feasibility studies and creating design solutions in CATIA V5 while managing parts in ENOVIA, influencing the final layout of the airline's A220 cabin.
- Developed internal tools using Google Sheets Apps Script to automate the tracking of aircraft cabin drawings aiding in production ramp-up and the reduction of HoV lead time from 12 to 9 months.
- Drafted multiple cabin configuration and monument drawings using CATIA V5 to support HoV engineering.

Delta Elevator Co Ltd

May - Aug. 2023

Mechanical Design Engineer

Kitchener, ON

- Developed a set of 3 adjustable oil buffer stands using Inventor sheet metal and weldment design to model and Ansys FEA to validate the design and chosen hardware thus eliminating the need for custom buffer stands.
- Designed mechanical joints for a traveling cable guard using DFMA and Lean Manufacturing principles minimizing installation hardware and prolonging the lifespan by 25% for this crucial component.

EDUCATION

University of Waterloo

Sept. 2022 - Apr. 2027

Bachelor of Applied Science, Honours Mechanical Engineering, Co-op (3.8/4.0 GPA)

Waterloo, ON